

AI Patient Follow-Up and Care Management Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives the patient's care plan, including the discharge date, follow-up appointment, prescribed tests, and medication schedule. It then begins monitoring the patient's scheduled care.
Actions	Send appointment/test reminders; send medication reminders; record the patient's response or confirmation; escalate an ignored reminder to a caregiver or doctor; update the patient's care record.
Transition Model	When the agent sends a reminder, receives a patient response, detects a missed item, or updates the record, the patient's care-plan state changes. For example: reminder pending → reminder sent → patient confirms, or reminder sent → no response → escalation.
Path Cost	The cost may be measured by the number of missed or unresolved care events, reminder attempts, or escalation actions. The objective is to keep the patient's care plan on schedule with timely intervention.
Goal Test	The goal is reached when scheduled follow-up appointments, tests, and medication doses are completed on time, with missed or unresolved items flagged early enough for caregiver or doctor intervention.
State Space	All possible combinations of appointment status, test status, medication status, reminder status, patient responses, and escalation status within the patient's care plan.
Search Space	All possible sequences of monitoring the care plan, sending reminders, receiving responses, recording completion, escalating missed items, and updating the care record.
Agents	Multiple agents. The intelligent monitoring agent operates with patients, caregivers, and doctors, who can also act within the patient-care environment.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent only knows information recorded in the system or confirmed by the patient. It cannot directly observe whether a medication dose was actually taken.
Determinism	Stochastic / Non-deterministic	Sending a reminder does not guarantee that the patient will respond or follow through. Human behaviour introduces uncertainty.
Episodes	Sequential	Earlier reminders, missed appointments, and patient responses affect later reminders and escalation decisions.
Change	Dynamic	Appointments can be rescheduled, new tests can be added, and patient responses can arrive while the agent is operating.
Values	Discrete	Appointments, medication doses, reminders, confirmations, and escalations are distinct countable events.
Agents	Multi-Agent	The monitoring agent operates in an environment involving patients, caregivers, and doctors, who can also take actions.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through many possible states: care plan received, reminders pending, reminders sent, patient response received, medication/test/appointment completed, missed item detected, escalation required, and updated care plan.

Final Answer: The AI Patient Follow-Up and Care Management Agent is best classified as a **Multi-state agent** because it passes through multiple changing states while monitoring the patient's care plan, sending reminders, recording responses, escalating missed

items, and updating the record. It is also **partially observable** and **stochastic** because it cannot directly observe every patient action and cannot guarantee that a patient will respond or follow the care plan.